Post-Lie Magnus expansion equals the weighted BCH-recursion at weight one (Theorem 5.3, the paper's main theorem)

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned source file sources/sigma-2022-023-post-lie-magnus-expansion-and-bch-recursion/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Printed numbers anchored to the published PDF (SIGMA 18 (2022) 023, stream-extracted text lane-s3/sig22-pdf-anchor-023-streams.txt, census pdf_anchor_evidence lane-s3/sig22-pdf-anchor-023-summary.json): Theorem 5.1 (T9) and Theorem 5.3 ('main Theorem') surface in the stream text ('Proof of Theorem 5.1', s021/s022); Proposition 5.2 is bracketed by the confirmed neighbours; the sectionwise shared master counter Theorem is declared in the author's newtheorem block (source lines 18-27: Corollary, Lemma, Proposition hang on the Theorem counter, the starred Theorem* at line 19, the definition-style group with its brace pair on lines 23 and 27).

Editorial note. Theorem-family numbers and equation numbers inside the card are locally renumbered (contexts taken from other sections of the source cannot keep their paper numbers under the sectionwise counters); every cross-reference is a symbolic ref and resolves inside the card; cite closure delivered as card-local bibliography fragments with href rendered plain if any (no hyperref in baseline); counters seeded from in-section source counts via the master counter name; author macro subset copied verbatim per CONTROL 8.8.165 (lines 11/41/42/46); no extra packages needed (wasysym omitted: no wasysym-only symbol in any delivered span).

Editorial note. Key-shape watch: the claim key of the main-theorem card carries the verbatim author label 'main Theorem' including its space (source line 674); this is deliberate and stream-anchored.

Editorial note. The counted span is the author Theorem environment with the verbatim spaced label 'main Theorem' (source lines 674-676) with its single proof environment (source lines 678-694).

Editorial note. The statement's \eqref{pl-Magnus} and \eqref{chi-lambda-rec} and the proof's \eqref{exp-moins-lambda-x}, \eqref{IPLME} and \eqref{ET9} all resolve through the delivered contexts; the context-span \cite keys (KLM06 at source line 416, ELM2015 at source line 522) are closed by the card-local bibliography fragment.

Editorial note. Printed numbers anchored to the published PDF (SIGMA 18 (2022) 023, stream-extracted text lane-s3/sig22-pdf-anchor-023-streams.txt, census pdf_anchor_evidence lane-s3/sig22-pdf-anchor-023-summary.json): Theorem 5.3 confirmed.

Editorial scope. Single environment card cut from the SIGMA article doi:10.3842/SIGMA.2022.023 (sigma-2022-023-post-lie-magnus-expansion-and-bch-recursion); statement and proof are the author's own text at source lines 674-676 and 678-694.

Allowed standard prerequisites (editorial).

- Weight-one specialisation of the inverse weighted recursion. Delivered in context (source lines 416-440): exp-moins-lambda-x with its KLM06 quotation frame.. Load-bearing: the proof opens from \eqref{exp-moins-lambda-x} specialised to lambda=1 (source line 679).
- Inverse post-Lie Magnus expansion IPLME. Delivered in context (source lines 554-559).. Load-bearing: combined with ET9 and res.2 at source line 689.
- Weight-one exponential factorization ET9 (Theorem 5.1). Delivered in context (source lines 605-611).. Load-bearing: \eqref{ET9} enters the res.3 combination (source line 689).

Theorem 5.3. *The post-Lie Magnus expansion χ, described in* (5.4)*, coincides with the weighted BCH-recursion χ_λ recursively given by* (5.1)*, with weight $\lambda = 1$.*

Proposition 5.4 ([1, Proposition 11])**.** *Let $\big(\mathcal{A}, \mathcal{R}\big)$ be a complete filtered Rota–Baxter algebra of weight $\lambda \neq 0$, and set $\widetilde{\mathcal{R}} := -\lambda \mathrm{id}_{\mathcal{A}} - \mathcal{R}$. The λ-weighted BCH-recursion is written*

$$\chi_\lambda(x) = x + \frac{1}{\lambda}\widetilde{\mathrm{BCH}}\big(\mathcal{R}(\chi_\lambda(x)), \widetilde{\mathcal{R}}(\chi_\lambda(x))\big), \tag{5.1}$$

for all $x \in \mathcal{A}_1$. It can be simplified to

$$\chi_\lambda(x) = x - \frac{1}{\lambda}\widetilde{\mathrm{BCH}}\big(-\mathcal{R}(\chi_\lambda(x)), \lambda x\big).$$

Its inverse is given by

$$\chi_\lambda^{-1}(x) = x - \frac{1}{\lambda}\widetilde{\mathrm{BCH}}\big(\mathcal{R}(x), \widetilde{\mathcal{R}}(x)\big).$$

Moreover, the factorization obtained in (5.3) *becomes*

$$\exp\big(\mathcal{R}(\chi_\lambda(x))\big)\exp\big(\widetilde{\mathcal{R}}(\chi_\lambda(x))\big) = \exp(-\lambda x). \tag{5.2}$$

$$\exp\big(\mathcal{R}(\chi(x))\big)\exp\big(\widetilde{\mathcal{R}}(\chi(x))\big) = \exp(x), \tag{5.3}$$

Both these exponential functions map $\mathrm{Prim}\big(F_E^{\mathrm{pl}}\big)$ bijectively onto $G\big(F_E^{\mathrm{pl}}\big)$. See [2] for details. The post-Lie Magnus expansion χ is the bijective map from $\widehat{\mathcal{L}\big(\mathcal{T}_E^{\mathrm{pl}}\big)}$ onto itself defined by

$$\exp^*\big(\chi(f)\big) = \exp(f),$$

namely,

$$\chi(f) = \log^*\big(\exp(f)\big). \tag{5.4}$$

5.1 Inverse post-Lie Magnus expansion

The inverse post-Lie Magnus expansion θ is the bijective map from $\widehat{\mathcal{L}\big(\mathcal{T}_E^{\mathrm{pl}}\big)}$ onto itself given by the following formula

$$\theta(f) = \log(\exp^*(f)) \tag{5.5}$$

Theorem 5.5. *Let $\big(\mathcal{A}, \mathcal{R}\big)$ be a complete filtered Rota–Baxter algebra of weight $\lambda = 1$. We have the following equality in $\widehat{\mathcal{U}(\mathcal{A})}$ for any $x \in \mathcal{A}$ and $t \in K$:*

$$\exp^*(tx) = \exp\big(-t\widetilde{\mathcal{R}}(x)\big) \exp\big(-t\mathcal{R}(x)\big), \tag{5.6}$$

where $\widetilde{\mathcal{R}} = -\mathrm{id}_{\mathcal{A}} - \mathcal{R}$, and $*$ is the associative product defined in (5.7)*, using the post-Lie product $x \rhd y = [\mathcal{R}(x), y]$.*

$$X * Y := X_{(1)}\big(X_{(2)} \rhd Y\big), \tag{5.7}$$

Proof. From equation (5.2), specialized to $\lambda = 1$, and by setting $\theta_{\mathrm{BCH}} := \chi_1^{-1}$, we obtain that

$$\exp\big(-\theta_{\mathrm{BCH}}(tx)\big) = \exp\big(t\mathcal{R}(x)\big) \exp\big(t\widetilde{\mathcal{R}}(x)\big)$$

in $\widehat{\mathcal{U}(\mathcal{A})}$, which is equivalent to

$$\exp\big(\theta_{\mathrm{BCH}}(tx)\big) = \exp\big(-t\widetilde{\mathcal{R}}(x)\big) \exp\big(-t\mathcal{R}(x)\big). \tag{5.8}$$

From (5.5), (5.6) and (5.8) we have

$$\exp\big(\theta_{\mathrm{BCH}}(tx)\big) = \exp^*(tx) = \exp\big(\theta(tx)\big). \tag{5.9}$$

Then the two θ's, namely the inverse BCH-recursion in (5.8) and the inverse post-Lie Magnus expansion (5.9), do coincide. □

References

[1] Ebrahimi-Fard K., Guo L., Manchon D., Birkhoff type decompositions and the Baker–Campbell–Hausdorff recursion, *Comm. Math. Phys.* **267** (2006), 821–845, arXiv:math-ph/0602004.

[2] Ebrahimi-Fard K., Lundervold A., Munthe-Kaas H.Z., On the Lie enveloping algebra of a post-Lie algebra, *J. Lie Theory* **25** (2015), 1139–1165, arXiv:1410.6350.